MILITARY SPECIFICATION

GLYCERYL TRIRICINOLEATE

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one grade of glyceryl triricinoleate.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables for Percent Defective

PUBLICATIONS

Naval Sea Systems Command (Code Ident 53711)

OD 18869	Density of Liquids, Determination of
OD 18879	Acid Number of Organic Materials, Determination of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: (Commander, Naval Air Systems Command, Engineering Division, Standardization Section, Code: AIR-52021, Department of the Navy, Washington, D.C. 20361) by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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OD 18882 Refractive Index of Liquids, Determination of

OD 18893 Water Content, Determination of

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids shall apply.

American Society for Testing and Materials (ASTM)

D 1959 Iodine Value of Drying Oils and Fatty Acids

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, Pa.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish a sample for first article inspection and approval (see 4.4 and 6.3).

3.2 Chemical and physical properties. The chemical and physical properties of the glyceryl triricinoleate shall be in accordance with table I.

TABLE I. Chemical and physical properties.

Property	Minimum	Maximum
Hydroxyl number, mg KOH/g	160	167
Water content, wt %	--	0.04
Acid number, mg KOH/g	--	0.8
Iodine number, cg/g	84	88
Density, 25°C, g/ml	0.954	0.958
Refractive index, $n_D^{25°C}$	1.4770	1.4785

3.3 Workmanship. The material shall be free from all impurities and other defects that could adversely affect its performance. The material shall be uniform in quality and shall be manufactured in accordance with standard manufacturing procedures of the industry.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling. Sampling shall be in accordance with MIL-STD-414, Inspection Level III, sections A and C, at the Acceptable Quality Levels (AQL) shown in table II for the following lower specification limits(LSL) and upper specification limits(USL).

TABLE II. Acceptable quality levels.

	AQL,%	
Property	LSL	USL
Hydroxyl number	1.0	1.0
Water content	--	1.0
Acid number	--	4.0
Iodine number	2.5	2.5
Density	1.5	1.5
Refractive index	1.5	1.5

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.4)
2. Quality conformance inspection (see 4.5)

4.4 First article inspection. First article inspection shall be performed by the contractor, after award of contract and prior to production, at a location acceptable to the Government. First article inspection shall be performed on samples which have been produced with equipment and procedures normally used in production. First article approval is valid only on the contract under which it is granted, unless extended by the Government to other contracts.

4.4.1 Sample size. The sample size shall be as specified in the contract or purchase order.

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4.4.2 Inspection routine. The sample shall be subjected to all of the quality conformance tests specified herein plus any additional tests specified by the procuring activity.

4.4.3 Failures. Failures in excess of those allowed in quality conformance inspection shall be cause for refusal to grant first article approval.

4.5 Quality conformance inspection. Quality conformance inspection shall be as specified in table III.

TABLE III. Quality conformance inspection.

Inspection	Test method
Hydroxyl number	see 4.6.1
Water content	OD 18893
Acid number	OD 18879
Iodine number	ASTM D 1959
Density	OD 18869
Refractive index	OD 18882
Visual inspection	see 4.6.2

4.6 Test methods.

4.6.1 Determination of hydroxyl number. Excess acetic anhydride-pyridine reagent is reacted with this organic hydroxyl compound to form esters of acetic acid. The excess pyridinium acetate is titrated with standard sodium hydroxide to the phenolphthalein endpoint. The difference in titer between the reagent standardization and sample is used to calculate the hydroxyl number of the sample.

(a) Weigh to the nearest milligram (mg) a sample of 1.8 to 2.2 grams into an Erlenmeyer flask equipped with a standard taper. All apparatus must be clean and dry before use.

(b) Pipet 50 milliliters (ml) of acetic anhydride-pyridine reagent into the reaction flask. Swirl and add a boiling chip. (4:100 acetic anhydride-pyridine concentration is prepared by adding in the proportion of 4 ml of chemically pure acetic anhydride to 100 ml reagent grade pyridine. The acetic anhydride-pyridine reagent darkens with age and should not be retained for more than 5 days. This solution should give a blank of 90 to 100 ml with 0.5N NaOH and may be adjusted if necessary.)

(c) Connect the flask to a water condenser and reflux using a hot plate or heating mantle for 120 minutes. The reflux period is started from the time when a drop falls every one or two seconds from the condenser tip inside the flask. Adjust the heat to maintain this rate.

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(d) After esterification add 10 ml of distilled water by slowly pouring down the condenser tube, being careful to avoid bumping. Hydrolyze for 5 minutes at reflux temperature.

(e) Raise the flask and condenser as a unit from the heat source and allow the condenser to drain for a few minutes. Remove the flask, stopper and cool to room temperature. Add 5 drops of 1 percent phenolphthalein in pyridine and titrate with standard 0.5N NaOH to the pink endpoint that remains for 30 seconds. (Prepare 1 percent phenolphthalein indicator by adding 1 gram of phenolphthalein to 100 ml of pyridine.)

(f) Run two reagent standardizations by using the same pipet as for the sample, then transfer the acetic anhydride-pyridine reagent into a clear 250-ml glass-stoppered Erlenmeyer flask, add 10 ml of water, and let stand at room temperature at least 5 minutes. Titrate with standard NaOH to the phenolphthalein endpoint.

(g) Determine the acid number of the sample in accordance with OD 18879.

(h) Calculate the hydroxyl number as follows:

$$\text{Hydroxyl number, mg KOH/g} = \frac{(B - S) \times N \times 56.1}{W} + A$$

Where: B= Volume NaOH used for blank, ml
S= Volume NaOH used for sample, ml
N= Normality of NaOH
W= Weight of the sample, g
A= Acid number, mg KOH/g

4.6.2 Visual inspection. The material shall be visually inspected for conformance to the requirements of 3.3 and section 5.

5. PACKAGING

5.1 Packing. The material shall be packed in clean, dry containers of the size and type specified by the procurement document. The packing container shall be so constructed as to insure acceptance by common or other carrier for safe transportation, at the lowest rate, to the point of delivery specified by the purchase order or contract.

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5.2 Marking. Containers shall be marked in accordance with MIL-STD-129. Marking shall include, but not be limited to, the following information:

(a) Number and date of this specification
(b) Manufacturer's and supplier's name(s)
(c) Material trade name and chemical name
(d) Lot number
(e) Date and place of manufacture
(f) Net weight

6. NOTES

6.1 Intended use. The material described herein is intended for use as an ingredient in solid rocket motors.

6.2 Ordering data. Procurement documents should specify, but not be limited to, the following:

(a) Title, number, and date of this specification
(b) Type and size of container desired
(c) Place of inspection
(d) Minimum lot size, if applicable
(e) Place of delivery
(f) Request for three copies of results of quality conformance inspection (if source inspected)
(g) First article sample size and any additional tests, if required

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation(ASPR). The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, test and approval of the first article.

6.4 Suggested source. The following is for informational purposes and is in no way restrictive. A satisfactory product has been obtained from Baker Castor Oil, 5585 61'st Street, Los Angeles, California.

Custodian:
Navy - AS

Preparing activity:
Navy - AS

Project No: 6810-NA58

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-G-85088(AS) Glyceryl Triricinoleate

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

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